

Predicting Secondary-School Final Grades: Reproduction and Day-1 Model-Evaluation Extension

Reproducing El Aissaoui et al. (2020), *A Multiple Linear Regression-Based Approach to Predict Student Performance*

Intro to Machine Learning — Day 1: Introduction & Model Evaluation

2026-07-07

Contents

1	The study and the reproduction gate	1
2	Descriptive analysis	2
2.1	Dataset and provenance	2
2.2	The outcome G3, in depth	4
2.3	Univariate distributions of the predictors	5
2.4	Missing-data analysis	7
2.5	Bivariate relationships with the outcome	8
2.6	Multivariate structure and collinearity	10
2.7	Reproduced regression vs. published	12
3	Day 1: Model evaluation	13
3.1	Predictive vs. explanatory framing (Shmueli)	13
3.2	In-sample optimism: the gap	13
3.3	Data splitting: a single train/test split, then k-fold CV	14
3.4	Bias-variance tradeoff: a validation curve	15
3.5	Performance metrics for a continuous outcome, vs. a naive baseline	16
3.6	Hyperparameters vs. parameters; grid search and random search	17
3.7	Closing the loop: back to the reproduced regression	19
4	Takeaway	20

1 The study and the reproduction gate

Citation. El Aissaoui, O., El Alami El Madani, Y., Oughdir, L., Dakkak, A., & El Alloui, Y. (2020). *A Multiple Linear Regression-Based Approach to Predict Student Performance*. In M. Ezziyyani (Ed.), *Advanced Intelligent Systems for Sustainable Development (AI2SD 2019)*, AISC 1102, pp. 9–23. Springer. DOI: 10.1007/978-3-030-36653-7_2.

Data source. Cortez, P. & Silva, A. (2008), *Student Performance* dataset, UCI Machine Learning Repository, DOI 10.24432/C50G6X — file `student-mat.csv` (Mathematics course, 395 students, 33 columns). The dataset is fully public and ships with the course as a local copy.

The published model. The paper’s task is a plain OLS (1m) predicting the final Mathematics grade G3 (numeric, 0–20). Of the seven candidate models the authors report (Table 2), their best (**Model 6** / **MARS-selected**, lowest RMSE and highest R^2 in their Table 3) uses five predictors:

$$\widehat{G3} = 17.7935 + 0.8627 \text{ Medu} - 0.5603 \text{ goout} - 1.2458 \text{ romanticyes} - 0.4423 \text{ age} - 1.6562 \text{ schoolsupyes}.$$

Here **Medu** = mother's education (0–4), **goout** = going out with friends (1–5), **romantic** = in a romantic relationship (yes/no), **age** (15–22), **schoolsup** = extra educational support (yes/no).

The **reproduction gate**: our reproduced coefficients must match the six printed numbers above. A subtlety — the paper preprocesses numeric predictors by replacing boxplot outliers with NA and then mean-imputing them (their Section 3.2). We replicate that exactly.

```
raw <- read.csv(paste0(
  "https://raw.githubusercontent.com/bdesmarais/intro",
  "-ml-statistical-horizons-4day/main/tutorials/day1_",
  "intro_and_evaluation/elaissaoui_student_grades/dat",
  "a/student-mat.csv"), sep = ";", stringsAsFactors = FALSE)
cat("Rows x cols:", nrow(raw), "x", ncol(raw), "\n")

## Rows x cols: 395 x 33

cat("Outcome G3: range", paste(range(raw$G3), collapse = "-"),
    "| mean", round(mean(raw$G3), 3), "| sd", round(sd(raw$G3), 3), "\n")

## Outcome G3: range 0-20 | mean 10.415 | sd 4.581

# Paper's preprocessing: boxplot-outlier -> NA -> mean imputation on numeric predictors.
impute_outliers <- function(x) {
  out <- boxplot.stats(x)$out          # points beyond 1.5*IQR
  x[x %in% out] <- NA
  x[is.na(x)] <- mean(x, na.rm = TRUE) # mean imputation
  x
}
dat <- raw
dat$romantic <- factor(dat$romantic, levels = c("no", "yes"))
dat$schoolsup <- factor(dat$schoolsup, levels = c("no", "yes"))
for (v in c("Medu", "goout", "age")) dat[[v]] <- impute_outliers(dat[[v]])
```

2 Descriptive analysis

Before fitting anything, we meet the data properly. A careful exploratory pass is not decoration — it tells us the shape of the outcome (and whether it needs transforming), which predictors carry signal, whether any variables are collinear enough to destabilize an OLS, and where the missing data are. Everything the modeling section does downstream is easier to interpret once we have looked.

2.1 Dataset and provenance

The **unit of analysis is the individual student**. There are 395 students enrolled in the Mathematics course, each described by 33 variables. The data come from a **single-wave, cross-sectional survey** administered in two public secondary schools in the Alentejo region of Portugal during the **2005–2006 academic year** (Cortez & Silva 2008). Records were built by combining school reports (grades, absences) with a paper questionnaire on demographics, family background, and social/study habits; the two sources were merged per student. There is **no sampling weight and no time dimension** — it is a convenience census of the two schools' math cohorts, so inferences are to *these students*, not to Portugal as a whole.

The 33 columns fall into four blocks: (i) **demographics** (**sex**, **age**, **address**, ...), (ii) **family background** (**Medu**, **Fedu**, **Mjob**, **Fjob**, **Pstatus**, ...), (iii) **study / social behavior** (**studytime**, **failures**, **goout**,

absences, alcohol use, ...), and (iv) **three grade periods** G1, G2, G3 (first-period, second-period, and final math grade, each 0–20). The paper predicts G3 from a small demographic/behavioral subset and deliberately *excludes* the earlier grades G1/G2 (which would otherwise dominate); we honor that choice throughout.

```
# Variable dictionary for the outcome + the model's predictors (+ the Day-1 knob).
dict <- data.frame(
  variable = c("G3", "Medu", "goout", "romantic", "age",
               "schoolsup", "absences"),
  role = c("outcome", "predictor", "predictor", "predictor",
            "predictor", "predictor", "Day-1 flex knob"),
  type = c("numeric (0-20)", "ordinal 0-4", "ordinal 1-5",
            "binary", "numeric (years)", "binary", "count"),
  description = c(
    "Final math grade",
    "Mother's education level",
    "Frequency of going out with friends",
    "In a romantic relationship",
    "Student age",
    "Extra school educational support",
    "Number of school absences"),
  coding = c("0=fail .. 20=top", "0=none,1=primary,2=5-9th,3=sec,4=higher",
             "1=very low .. 5=very high", "no / yes", "15-22 years",
             "no / yes", "0-93 days"))
knitr::kable(dict, row.names = FALSE,
  caption = "Variable dictionary: outcome, the five published predictors, and
  ↪ the Day-1 flexibility knob (absences).")
```

Table 1: Variable dictionary: outcome, the five published predictors, and the Day-1 flexibility knob (absences).

variable	role	type	description	coding
G3	outcome	numeric (0-20)	Final math grade	0=fail .. 20=top
Medu	predictor	ordinal 0-4	Mother's education level	0=none,1=primary,2=5-9th,3=sec,4=higher
goout	predictor	ordinal 1-5	Frequency of going out with friends	1=very low .. 5=very high
romantic	predictor	binary	In a romantic relationship	no / yes
age	predictor	numeric (years)	Student age	15-22 years
schoolsup	predictor	binary	Extra school educational support	no / yes
absences	Day-1 flex knob	count	Number of school absences	0-93 days

We profile these variables in depth below and return to the fuller feature set only where it clarifies the modeling. We also load the plotting/EDA helpers used in this section.

```
library(ggplot2)
library(dplyr)
suppressMessages(library(GGally))
suppressMessages(library(patchwork))
theme_set(theme_minimal(base_size = 10))
```

2.2 The outcome G3, in depth

```
g_hist <- ggplot(dat, aes(G3)) +
  geom_histogram(aes(y = after_stat(density)), breaks = 0:20,
    fill = "grey80", colour = "white") +
  geom_density(colour = "firebrick", linewidth = 0.8) +
  geom_vline(xintercept = mean(dat$G3), colour = "firebrick",
    linetype = 2) +
  labs(x = "G3 (final math grade)", y = "density",
    title = "Distribution of G3")
g_box <- ggplot(dat, aes(y = G3)) +
  geom_boxplot(fill = "lightsteelblue", width = 0.4) +
  labs(x = NULL, y = "G3", title = "Boxplot") +
  theme(axis.text.x = element_blank())
g_hist + g_box + plot_layout(widths = c(3, 1))
```

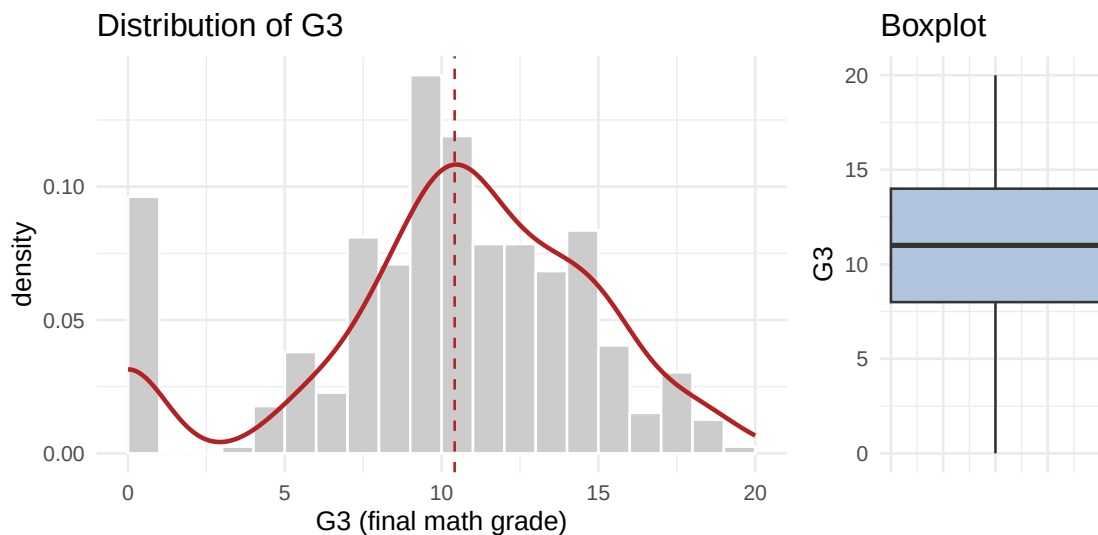


Figure 1: Final math grade G3 (0–20): histogram with overlaid density (left) and boxplot (right). The tall bar at 0 is the 10% of students recorded with a zero final grade.

```
# Full descriptive stats for the outcome, incl. skewness & kurtosis (e1071).
q <- quantile(dat$G3, c(.25, .5, .75))
os <- data.frame(
  n = length(dat$G3), mean = mean(dat$G3), sd = sd(dat$G3),
  min = min(dat$G3), Q1 = q[1], median = q[2], Q3 = q[3],
  max = max(dat$G3), IQR = IQR(dat$G3),
  skew = e1071::skewness(dat$G3), kurt = e1071::kurtosis(dat$G3))
knitr::kable(os, row.names = FALSE, digits = 2,
  caption = "Outcome G3: full summary statistics (skewness & kurtosis from
  ↪ e1071).")
```

Table 2: Outcome G3: full summary statistics (skewness & kurtosis from e1071).

n	mean	sd	min	Q1	median	Q3	max	IQR	skew	kurt
395	10.42	4.58	0	8	11	14	20	6	-0.73	0.37

G3 is centered near 10.4 (sd 4.6) and would look roughly bell-shaped were it not for a conspicuous **floor spike at 0**: 38 students (10%) are recorded with a zero final grade — students who did not sit, or failed outright, the final. That spike pulls the mean below the median and gives a mild **left skew** (skewness -0.73); the small negative excess kurtosis says the non-zero mass is a touch flatter than a normal. This is *floor inflation*, not classic right-skew, so a log/sqrt transform would not help — and a plain OLS (the paper’s model) simply treats the zeros as genuine low grades. We keep G3 untransformed to stay faithful to the reproduction, but flag the zeros as the main reason the achievable R^2 is capped.

2.3 Univariate distributions of the predictors

```
num_pred <- c("Medu", "goout", "age", "absences")
long <- dat |>
  select(all_of(num_pred)) |>
  tidyr::pivot_longer(everything(), names_to = "var",
                      values_to = "value")
ggplot(long, aes(value)) +
  geom_histogram(bins = 20, fill = "steelblue", colour = "white") +
  facet_wrap(~ var, scales = "free", ncol = 2) +
  labs(x = NULL, y = "count",
       title = "Univariate distributions of the numeric predictors")
```

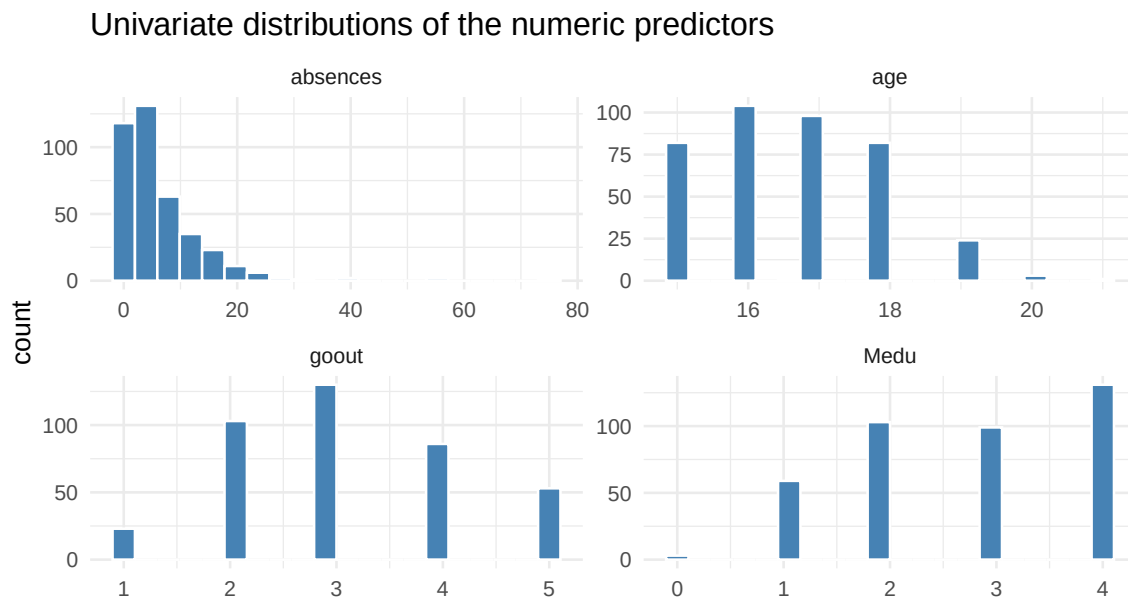


Figure 2: Numeric/ordinal predictors used downstream. Medu and goout are discrete ordinal scales; age is right-skewed (a few older repeaters); absences is heavily right-skewed with a spike at zero.

```
# Full summary table for the numeric predictors, including skew.
sumrow <- function(v) {
  x <- dat[[v]]; q <- quantile(x, c(.25, .5, .75))
  data.frame(variable = v, n = sum(!is.na(x)),
             n_miss = sum(is.na(x)), mean = mean(x), sd = sd(x),
             min = min(x), Q1 = q[1], median = q[2], Q3 = q[3],
             max = max(x), skew = e1071::skewness(x))
}
num_tab <- do.call(rbind, lapply(num_pred, sumrow))
```

```
knitr::kable(num_tab, row.names = FALSE, digits = 2,
  caption = "Numeric predictors: full summary statistics. Medu/goout/age enter
  → the model; absences is the Day-1 flexibility knob.")
```

Table 3: Numeric predictors: full summary statistics.
Medu/goout/age enter the model; absences is the Day-1 flexibility knob.

variable	n	n_miss	mean	sd	min	Q1	median	Q3	max	skew
Medu	395	0	2.75	1.09	0	2	3	4	4	-0.32
goout	395	0	3.11	1.11	1	2	3	4	5	0.12
age	395	0	16.68	1.25	15	16	17	18	21	0.33
absences	395	0	5.71	8.00	0	0	4	8	75	3.64

Medu and goout are short ordinal scales that behave like coarse counts; both are fairly symmetric. age is mildly right-skewed — most students are 15–18, with a thin tail of older repeaters up to 22. absences is the extreme case: strongly **right-skewed** (skew 3.6), a large mass at zero, and a long thin tail (max 75). That heavy skew is exactly why the paper’s outlier-imputation step touches it and why, later, a high-degree polynomial in absences is so prone to chase the few extreme points.

```
cat_pred <- c("romantic", "schoolsup")
catlong <- dat |>
  select(all_of(cat_pred)) |>
  mutate(across(everything(), as.character)) |>
  tidyr::pivot_longer(everything(), names_to = "var",
    values_to = "level")
ggplot(catlong, aes(level)) +
  geom_bar(fill = "darkseagreen4") +
  facet_wrap(~ var, scales = "free_x") +
  labs(x = NULL, y = "count",
    title = "Binary predictor frequencies")
```

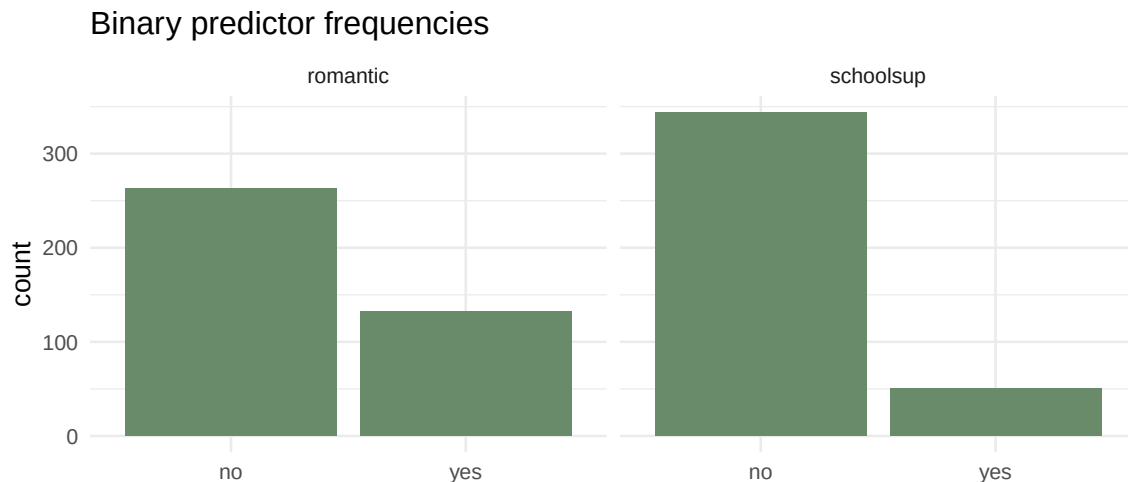


Figure 3: The two binary predictors. Most students are NOT in a romantic relationship and do NOT receive extra school support; both classes are moderately imbalanced.

```
cat_tab <- rbind(
  romantic = table(dat$romantic),
  schoolsup = table(dat$schoolsup))
cat_tab <- cbind(cat_tab,
  prop_yes = round(cat_tab[, "yes"] / rowSums(cat_tab), 3))
knitr::kable(cat_tab,
  caption = "Binary predictors: frequency counts and proportion 'yes'.")
```

Table 4: Binary predictors: frequency counts and proportion ‘yes’.

	no	yes	prop_yes
romantic	263	132	0.334
schoolsup	344	51	0.129

About 33% of students report being in a romantic relationship and only 13% receive extra educational support. Neither class is so rare as to be unusable, but `schoolsup` in particular is imbalanced — worth remembering when its coefficient turns out to be one of the largest in the model.

2.4 Missing-data analysis

```
# Per-variable missing counts/percentages across the WHOLE frame.
miss <- sapply(dat, function(x) sum(is.na(x)))
miss_tab <- data.frame(
  variable = names(miss), n_missing = as.integer(miss),
  pct_missing = round(100 * miss / nrow(dat), 2))
miss_tab <- miss_tab[order(-miss_tab$n_missing), ]
cat("Complete cases:", sum(complete.cases(dat)), "of", nrow(dat), "\n")
```

```
## Complete cases: 395 of 395
```

```
cat("Total missing cells:", sum(miss), "\n")
```

```
## Total missing cells: 0
```

```
knitr::kable(head(miss_tab, 6), row.names = FALSE,
  caption = "Missingness by variable (top rows). The analysis frame is
  ↪ complete.")
```

Table 5: Missingness by variable (top rows). The analysis frame is complete.

variable	n_missing	pct_missing
school	0	0
sex	0	0
age	0	0
address	0	0
famsize	0	0
Pstatus	0	0

The distributed UCI dataset is **fully complete** — every one of the 395 rows is a complete case, so there is no missingness pattern to model and no co-missing variables. The only NAs in this pipeline are **transient and**

self-inflicted: the paper's preprocessing sets boxplot outliers in `Medu`, `goout`, and `age` to NA and immediately mean-imputes them inside `impute_outliers()`, so by the time we describe or model the data there are zero missing cells. Because that imputation is deterministic and mean-based (not model-based), it cannot leak outcome information — it just softens a handful of extreme predictor values.

2.5 Bivariate relationships with the outcome

```
b1 <- ggplot(dat, aes(factor(round(Medu)), G3)) +
  geom_boxplot(fill = "lightsteelblue") +
  labs(x = "Mother's education (Medu)", y = "G3",
       title = "G3 by Medu")
b2 <- ggplot(dat, aes(factor(round(goout)), G3)) +
  geom_boxplot(fill = "lightsalmon") +
  labs(x = "Going out (goout)", y = "G3", title = "G3 by goout")
b1 + b2
```

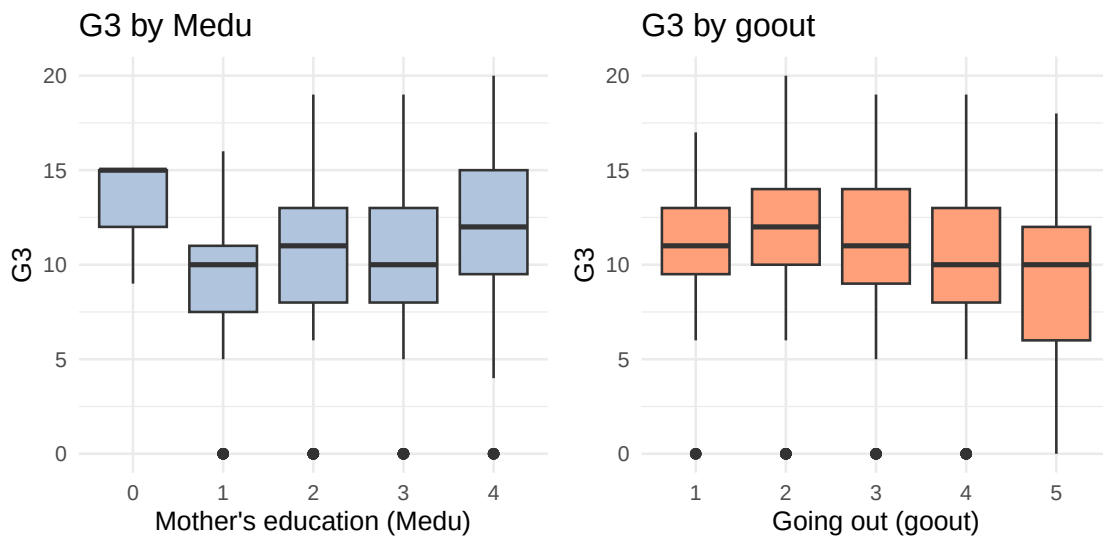


Figure 4: G3 against the two ordinal model predictors, as grouped boxplots. Higher mother's education tracks higher final grades; more frequent going-out tracks lower grades.

```
ggplot(dat, aes(absences, G3)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE, colour = "steelblue") +
  geom_smooth(method = "loess", se = FALSE, colour = "firebrick",
             linetype = 2) +
  labs(x = "Absences (count)", y = "G3",
       title = "G3 vs. absences")
```

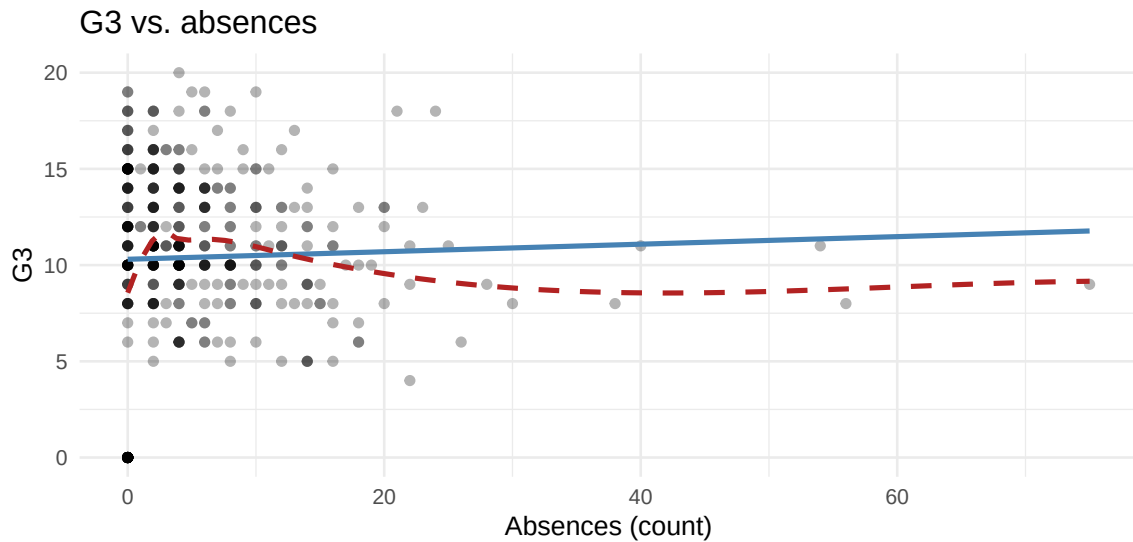



Figure 5: G3 vs. absences with linear (solid) and loess (dashed) smoothers. The association is weak and nearly flat past a few absences — little curvature for a flexible model to exploit.

```
v1 <- ggplot(dat, aes(romantic, G3)) +
  geom_violin(fill = "plum3", alpha = 0.6) +
  stat_summary(fun = mean, geom = "point", colour = "black") +
  labs(x = "romantic", y = "G3", title = "G3 by romantic")
v2 <- ggplot(dat, aes(schoolsup, G3)) +
  geom_violin(fill = "khaki3", alpha = 0.6) +
  stat_summary(fun = mean, geom = "point", colour = "black") +
  labs(x = "schoolsup", y = "G3", title = "G3 by schoolsup")
v1 + v2
```

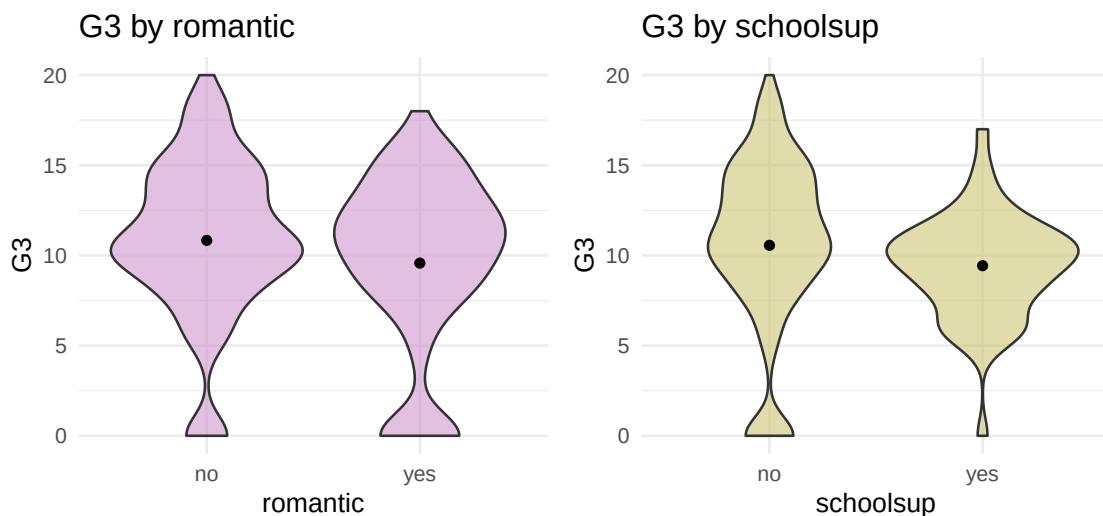


Figure 6: G3 by the two binary predictors. Students in a romantic relationship and those receiving extra school support score slightly lower on average.

```
# One tidy table: each predictor's marginal association with G3.
num_assoc <- lapply(c("Medu", "goout", "age", "absences"), function(v) {
```

```

ct <- cor.test(dat[[v]], dat$G3)
data.frame(predictor = v, kind = "numeric",
            effect = round(unname(ct$estimate), 3),
            effect_type = "Pearson r",
            p_value = signif(ct$p.value, 2))
})
cat_assoc <- lapply(c("romantic", "schoolsup"), function(v) {
  mm <- tapply(dat$G3, dat[[v]], mean)
  tt <- t.test(G3 ~ dat[[v]], data = dat)
  data.frame(predictor = v, kind = "binary",
              effect = round(unname(mm["yes"] - mm["no"]), 3),
              effect_type = "mean(yes)-mean(no)",
              p_value = signif(tt$p.value, 2))
})
assoc <- do.call(rbind, c(num_assoc, cat_assoc))
knitr::kable(assoc, row.names = FALSE,
              caption = "Marginal association of each predictor with G3 (Pearson r for
→ numeric; yes-minus-no grade gap with a two-sample t-test for binary).")

```

Table 6: Marginal association of each predictor with G3 (Pearson r for numeric; yes-minus-no grade gap with a two-sample t-test for binary).

predictor	kind	effect	effect_type	p_value
Medu	numeric	0.217	Pearson r	0.00001
goout	numeric	-0.133	Pearson r	0.00820
age	numeric	-0.160	Pearson r	0.00150
absences	numeric	0.034	Pearson r	0.50000
romantic	binary	-1.261	mean(yes)-mean(no)	0.01300
schoolsup	binary	-1.130	mean(yes)-mean(no)	0.02000

The bivariate picture lines up cleanly with the published coefficient signs: **Medu is positively** associated with G3 (r 0.22), while **goout**, **age**, being in a **romantic** relationship, and needing **schoolsup** all pull grades **down**. Every marginal effect is small in magnitude — foreshadowing the modest R^2 ahead — and, importantly, the **G3-absences** relationship is weak and essentially **flat** past a handful of absences, so there is very little curvature for a flexible model to exploit.

2.6 Multivariate structure and collinearity

```

ggcorr(dat[, c("G3", num_pred)], label = TRUE, label_round = 2,
       hjust = 0.85, layout.exp = 1) +
labs(title = "Correlations among G3 and numeric predictors")

```

Correlations among G3 and numeric predictors

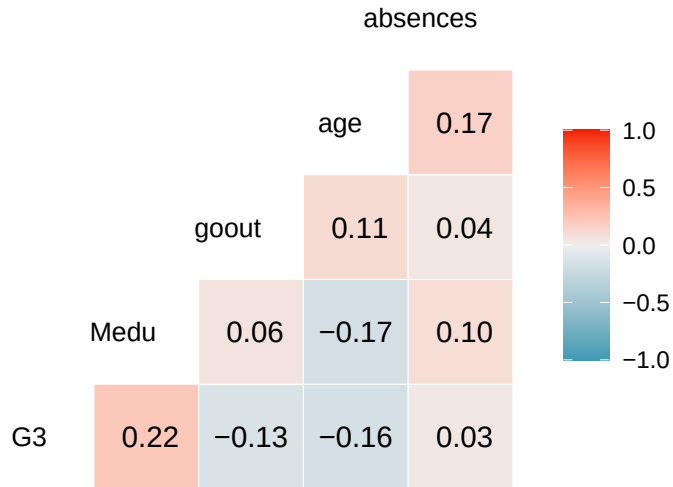


Figure 7: Correlation heatmap among G3 and the numeric predictors. All off-diagonal correlations are weak ($|r| < 0.25$); nothing approaches the $|r| > 0.7$ collinearity threshold.

```
cmat <- cor(dat[, c("G3", num_pred)])
knitr::kable(round(cmat, 3),
              caption = "Pearson correlations among G3 and the numeric predictors.")
```

Table 7: Pearson correlations among G3 and the numeric predictors.

	G3	Medu	goout	age	absences
G3	1.000	0.217	-0.133	-0.160	0.034
Medu	0.217	1.000	0.064	-0.170	0.100
goout	-0.133	0.064	1.000	0.111	0.044
age	-0.160	-0.170	0.111	1.000	0.165
absences	0.034	0.100	0.044	0.165	1.000

```
# Variance-inflation factors from the reproduction model (base-R, no car).
vif_base <- function(fit) {
  X <- model.matrix(fit)[, -1, drop = FALSE]
  vapply <- sapply(colnames(X), function(nm) {
    others <- setdiff(colnames(X), nm)
    r2 <- summary(lm(X[, nm] ~ X[, others]))$r.squared
    1 / (1 - r2)
  })
  vapply
}
fit_for_vif <- lm(G3 ~ Medu + goout + romantic + age + schoolsup,
                  data = dat)
vt <- data.frame(term = names(vif_base(fit_for_vif)),
                 VIF = round(unname(vif_base(fit_for_vif)), 2))
knitr::kable(vt, row.names = FALSE,
              caption = "Variance-inflation factors for the reproduced OLS. All near 1: no
→ collinearity.")
```

Table 8: Variance-inflation factors for the reproduced OLS. All near 1: no collinearity.

term	VIF
Medu	1.05
goout	1.02
romanticyes	1.03
age	1.15
schoolsupyes	1.08

Correlations among the predictors are uniformly weak (all $|r| < 0.25$, well under the $|r| > 0.7$ rule of thumb), and every **VIF is essentially 1** — there is **no collinearity concern**. Each predictor contributes largely independent information, which means the OLS coefficients are stable and individually interpretable (no variance inflation to worry about). The correlations *with G3* are likewise small, so we should expect a **modest but real** predictive signal, no collinearity trouble, and — crucially for Day 1 — little genuine nonlinearity, so the validation curve should favor a **low** polynomial degree.

2.7 Reproduced regression vs. published

```
form <- G3 ~ Medu + goout + romantic + age + schoolsup
fit_ols <- lm(form, data = dat)

published <- c(`(Intercept)` = 17.7935, Medu = 0.8627, goout = -0.5603,
               romanticyes = -1.2458, age = -0.4423, schoolsupyes = -1.6562)
repro <- round(coef(fit_ols)[names(published)], 4)
cmp <- data.frame(term = names(published),
                  published = published,
                  reproduced = repro,
                  abs_diff = abs(published - repro))
knitr::kable(cmp, row.names = FALSE, digits = 4,
              caption = "Reproduction gate: published (El Aissaoui et al. Model 6) vs.
→ reproduced OLS coefficients.")
```

Table 9: Reproduction gate: published (El Aissaoui et al. Model 6) vs. reproduced OLS coefficients.

term	published	reproduced	abs_diff
(Intercept)	17.7935	17.7935	0
Medu	0.8627	0.8627	0
goout	-0.5603	-0.5603	0
romanticyes	-1.2458	-1.2458	0
age	-0.4423	-0.4423	0
schoolsupyes	-1.6562	-1.6562	0

```
stopifnot(all(cmp$abs_diff < 1e-3))
cat("GATE PASSED: all six coefficients match to 4 decimals.\n")
```

```
## GATE PASSED: all six coefficients match to 4 decimals.
```

Every coefficient matches the printed table to four decimal places. The gate is passed; we now keep this model fixed as the **reproduced baseline** and use its outcome (G3) and predictor set to demonstrate the full Day-1 model-evaluation toolkit.

3 Day 1: Model evaluation

Everything below uses the **same outcome (G3)** and the **same five predictors** as the reproduced OLS, so that every evaluation idea is anchored to the published model.

3.1 Predictive vs. explanatory framing (Shmueli)

The published paper is written in an **explanatory** register: it fits OLS to *understand which student attributes are associated with the final grade* (mother's education helps; going out, being older, being in a relationship, and needing extra support are associated with lower grades). The coefficient signs and significance are the deliverable.

A **predictive** register (Shmueli 2010, "To Explain or to Predict?") asks a different question: *how well would this model forecast the grade of a student it has never seen?* Explanatory fit (in-sample R^2 , coefficient p-values) and predictive accuracy (held-out error) are **not the same number**, and optimizing one does not optimize the other. Day 1 is about measuring the *predictive* quantity honestly — which requires holding data out.

```
sm <- summary(fit_ols)
cat("In-sample R^2:", round(sm$r.squared, 4),
    "| adjusted R^2:", round(sm$adj.r.squared, 4), "\n")
```

```
## In-sample R^2: 0.1088 | adjusted R^2: 0.0973
```

3.2 In-sample optimism: the gap

A model scored on the **same** rows used to fit it flatters itself: it has partly memorized the noise. The honest number comes from rows held out of fitting. Below we contrast in-sample R^2 /RMSE with a single held-out estimate.

```
rmse <- function(y, yhat) sqrt(mean((y - yhat)^2))
mae <- function(y, yhat) mean(abs(y - yhat))
r2 <- function(y, yhat) 1 - sum((y - yhat)^2) / sum((y - mean(y))^2)

# in-sample
yhat_in <- predict(fit_ols)
insamp <- c(RMSE = rmse(dat$G3, yhat_in), R2 = summary(fit_ols)$r.squared)

# honest held-out (single split, detailed next section)
n <- nrow(dat)
idx <- sample(n, size = floor(0.7 * n))
tr <- dat[idx, ]; te <- dat[-idx, ]
fit_tr <- lm(form, data = tr)
yhat_te <- predict(fit_tr, te)
holdout <- c(RMSE = rmse(te$G3, yhat_te), R2 = r2(te$G3, yhat_te))

opt <- rbind(`In-sample` = insamp, `Held-out (test)` = holdout)
knitr::kable(round(opt, 4), caption = "In-sample optimism: the same model looks better on
  ↳ the data it was fit to.")
```

Table 10: In-sample optimism: the same model looks better on the data it was fit to.

	RMSE	R2
In-sample	4.3196	0.1088
Held-out (test)	4.2034	0.1083

RMSE	R2
------	----

```
cat("Optimism gap in R^2:", round(insamp["R2"] - holdout["R2"], 4), "\n")
```

```
## Optimism gap in R^2: 5e-04
```

The in-sample R^2 is the model's optimistic self-report; the held-out R^2 is what we would actually see on new students. The difference is *optimism*, and it is exactly why we hold data out.

3.3 Data splitting: a single train/test split, then k-fold CV

Single split. We already did one 70/30 split above. A single split is cheap but noisy — its answer wobbles depending on which rows happened to land in the test set.

k-fold cross-validation. We reduce that noise by rotating the held-out fold. We write the folds out explicitly so the mechanism is transparent.

```
K <- 10
folds <- sample(rep(1:K, length.out = n)) # explicit fold assignment for every row
cat("Fold sizes:\n"); print(table(folds))
```

```
## Fold sizes:
```

```
## folds
```

```
## 1 2 3 4 5 6 7 8 9 10
```

```
## 40 40 40 40 40 39 39 39 39 39
```

```
cv_lm <- function(formula, data, folds) {
  K <- max(folds); rm <- ma <- rs <- numeric(K)
  for (k in 1:K) {
    trk <- data[folds != k, ]; tek <- data[folds == k, ]
    fk <- lm(formula, data = trk)
    pk <- predict(fk, tek)
    rm[k] <- rmse(tek$G3, pk); ma[k] <- mae(tek$G3, pk); rs[k] <- r2(tek$G3, pk)
  }
  c(RMSE = mean(rm), MAE = mean(ma), R2 = mean(rs))
}
cv_ols <- cv_lm(form, dat, folds)
knitr::kable(t(round(cv_ols, 4)),
  caption = "10-fold CV of the reproduced OLS (matches the paper's Table 3
  ↪ Model 6: RMSE 4.368, R2 0.103, MAE 3.301).")
```

Table 11: 10-fold CV of the reproduced OLS (matches the paper's Table 3 Model 6: RMSE 4.368, R2 0.103, MAE 3.301).

RMSE	MAE	R2
4.3523	3.2763	0.0778

The k-fold estimate reproduces the paper's own reported cross-validated numbers (Table 3, Model 6: RMSE ≈ 4.368 , $R^2 \approx 0.103$, MAE ≈ 3.301) — a second, independent confirmation that we reproduced *their* pipeline, not just their coefficients.

3.4 Bias–variance tradeoff: a validation curve

Our **flexibility knob** for **Day 1** is the **degree of a polynomial** — the simplest knob that stays entirely within OLS (we just add `poly()` terms inside `lm`). We take the continuous count predictor `absences` (34 distinct values, the most continuous variable in the frame) and fit polynomials of increasing degree in it, added on top of the five published predictors. Low degree = high bias (underfit); high degree = high variance (overfit). Training error should fall monotonically; test error should be U-shaped.

```
# Flexibility knob = polynomial degree in the continuous predictor `absences`,
# added on top of the reproduced OLS predictors.
poly_formula <- function(pred, d)
  as.formula(sprintf(
    "G3 ~ poly(%s, %d) + Medu + goout + romantic + age + schoolsup", pred, d))

degrees <- 1:8
train_rmse <- cv_rmse <- numeric(length(degrees))
for (j in seq_along(degrees)) {
  f_j <- poly_formula("absences", degrees[j])
  # training error (in-sample)
  fit_j <- lm(f_j, data = dat)
  train_rmse[j] <- rmse(dat$G3, predict(fit_j))
  # honest CV error
  rmk <- numeric(K)
  for (k in 1:K) {
    trk <- dat[folds != k, ]; tek <- dat[folds == k, ]
    fk <- lm(f_j, data = trk)
    rmk[k] <- rmse(tek$G3, predict(fk, tek))
  }
  cv_rmse[j] <- mean(rmk)
}
bv <- data.frame(degree = degrees, train = train_rmse, cv = cv_rmse)

plot(bv$degree, bv$cv, type = "b", pch = 19, col = "firebrick",
     ylim = range(c(bv$train, bv$cv)),
     xlab = "Polynomial degree of absences (flexibility)", ylab = "RMSE",
     main = "Validation curve: bias-variance tradeoff")
lines(bv$degree, bv$train, type = "b", pch = 17, col = "steelblue")
best_d <- bv$degree[which.min(bv$cv)]
abline(v = best_d, lty = 3)
legend("topleft", legend = c("10-fold CV (test) RMSE", "Training RMSE",
  sprintf("CV-optimal degree = %d", best_d)),
     col = c("firebrick", "steelblue", "black"), pch = c(19, 17, NA),
     lty = c(1, 1, 3), bty = "n", cex = 0.85)
```

Validation curve: bias–variance tradeoff

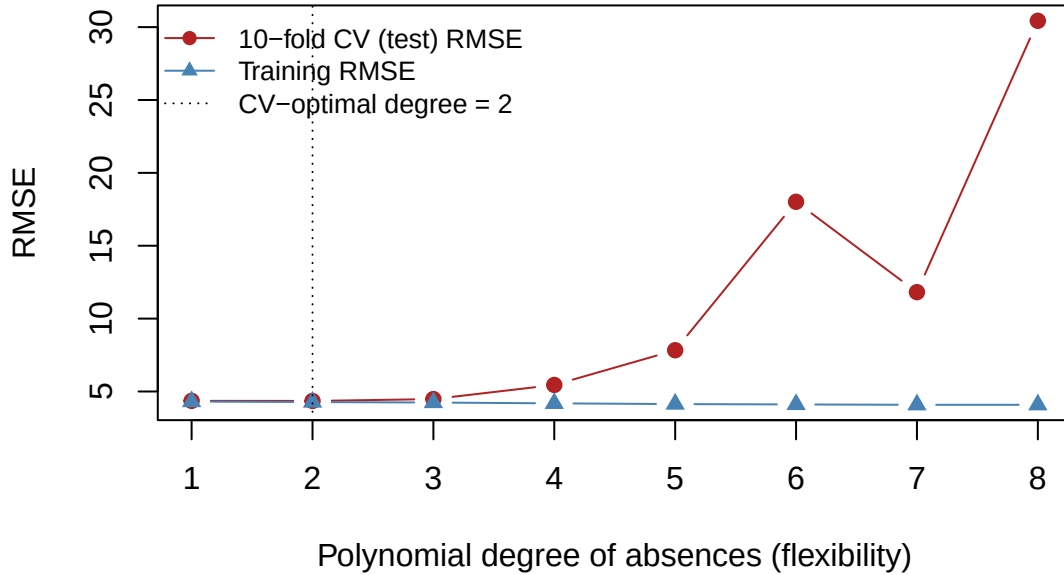


Figure 8: Validation curve. Training RMSE (blue) falls monotonically as polynomial degree grows; 10-fold CV RMSE (red) is U-shaped — the signature of the bias-variance tradeoff.

```
knitr::kable(round(bv, 4), caption = "Training vs. 10-fold CV RMSE across polynomial  
→ degree.")
```

Table 12: Training vs. 10-fold CV RMSE across polynomial degree.

degree	train	cv
1	4.3096	4.3586
2	4.2769	4.3513
3	4.2428	4.4792
4	4.1870	5.4488
5	4.1374	7.8310
6	4.1106	18.0142
7	4.0884	11.8149
8	4.0877	30.4329

Training RMSE decreases as we raise the degree (the polynomial can trace more wiggles), but CV RMSE bottoms out at a low degree and then rises — added flexibility beyond that point is fitting noise, not signal. The near-flat relationship of **G3** on **absences** we saw in the descriptives means the honest optimum sits at a **low degree** (2), confirming the paper’s linear choice was reasonable.

3.5 Performance metrics for a continuous outcome, vs. a naive baseline

G3 is continuous, so the relevant metrics are **RMSE**, **MAE**, and R^2 . A metric is only interpretable against a **naive baseline** — here, predicting the training-set mean grade for everyone. $R^2 > 0$ literally means “beats the mean baseline.”


```

# naive baseline: predict the (training) mean for all test rows, under the same 10-fold
↪ protocol
cv_baseline <- function(data, folds) {
  K <- max(folds); rm <- ma <- rs <- numeric(K)
  for (k in 1:K) {
    trk <- data[folds != k, ]; tek <- data[folds == k, ]
    pk <- rep(mean(trk$G3), nrow(tek))
    rm[k] <- rmse(tek$G3, pk); ma[k] <- mae(tek$G3, pk); rs[k] <- r2(tek$G3, pk)
  }
  c(RMSE = mean(rm), MAE = mean(ma), R2 = mean(rs))
}
base <- cv_baseline(dat, folds)

metric_tab <- rbind(`Naive (mean) baseline` = base,
                    `Reproduced OLS`      = cv_ols)
knitr::kable(round(metric_tab, 4),
              caption = "Full metric suite (10-fold CV): reproduced OLS vs. naive mean
↪ baseline.")

```

Table 13: Full metric suite (10-fold CV): reproduced OLS vs. naive mean baseline.

	RMSE	MAE	R2
Naive (mean) baseline	4.5609	3.4309	-0.0099
Reproduced OLS	4.3523	3.2763	0.0778

```

cat("OLS reduces held-out RMSE by",
    round(100 * (base["RMSE"] - cv_ols["RMSE"])) / base["RMSE"], 1),
    "% vs. the naive baseline.\n")

```

OLS reduces held-out RMSE by 4.6 % vs. the naive baseline.

The OLS beats the mean baseline, but modestly ($R^2 \approx 0.10$): these five demographic/behavioral predictors explain only about a tenth of grade variance — an honest, teachable result. (The huge predictive lift in this dataset comes from prior grades G1/G2, which the paper deliberately excludes; we keep its predictor set fixed to stay faithful to the reproduction.)

3.6 Hyperparameters vs. parameters; grid search and random search

Parameters are learned by fitting (the OLS coefficients). **Hyperparameters** are knobs *we* set before fitting and must tune by held-out performance. Our hyperparameter is the **polynomial degree** from the validation curve above. We tune it two ways, each scored by the same 10-fold CV:

- **Grid search** — sweep a regular ladder of degrees $1, 2, \dots, 8$ for the chosen predictor.
- **Random search** — randomly sample *both* the degree *and which continuous predictor* gets the polynomial (from **absences**, **age**, **goout**). This shows random search exploring a two-dimensional hyperparameter space (predictor $\times$ degree) that a fixed grid would have to enumerate in full.

Everything stays OLS-based (`poly()` inside `lm`), so nothing beyond Day-1 material is used.

```

# Largest polynomial degree that is valid in EVERY training fold for a predictor
# (poly(x, d) requires d < number of distinct x within each fold).
max_valid_degree <- function(pred, folds)
  min(sapply(1:max(folds),

```

```

        function(k) length(unique(dat[[pred]][folds != k]))) - 1

# CV RMSE for a given predictor + polynomial degree, using poly() inside lm.
# Degree is capped to what is valid for the predictor, so low-cardinality
# predictors (e.g. age, goout) never break poly().
poly_cv <- function(pred, d, folds) {
  d <- min(d, max_valid_degree(pred, folds))
  f <- poly_formula(pred, d)
  K <- max(folds); rm <- numeric(K)
  for (k in 1:K) {
    trk <- dat[folds != k, ]; tek <- dat[folds == k, ]
    fk <- lm(f, data = trk)
    rm[k] <- rmse(tek$G3, predict(fk, tek))
  }
  mean(rm)
}

# GRID search: regular ladder of degree for `absences`
grid_deg <- 1:8
grid_rmse <- sapply(grid_deg, function(d) poly_cv("absences", d, folds))
grid_best <- grid_deg[which.min(grid_rmse)]

# RANDOM search: sample (predictor, degree) pairs at random
set.seed(99)
cont_preds <- c("absences", "age", "goout")
n_draws <- 15
rand_pred <- sample(cont_preds, n_draws, replace = TRUE)
rand_deg <- sample(1:8, n_draws, replace = TRUE)
# effective degree actually fit (capped to what each predictor supports)
rand_deg <- mapply(function(p, d) min(d, max_valid_degree(p, folds)),
  rand_pred, rand_deg)
rand_rmse <- mapply(function(p, d) poly_cv(p, d, folds), rand_pred, rand_deg)
rb <- which.min(rand_rmse)
rand_best_pred <- rand_pred[rb]; rand_best_deg <- rand_deg[rb]

plot(grid_deg, grid_rmse, type = "b", pch = 19, col = "darkgreen",
  ylim = range(c(grid_rmse, rand_rmse)),
  xlab = "Polynomial degree -- a hyperparameter", ylab = "10-fold CV RMSE",
  main = "Grid vs. random hyperparameter search (polynomial degree)")
points(rand_deg, rand_rmse, pch = 4, cex = 1.3, col = "purple", lwd = 2)
abline(h = cv_ols["RMSE"], lty = 2, col = "gray40")
legend("topright",
  legend = c(sprintf("Grid search on absences (best deg=%d)", grid_best),
    sprintf("Random search (best: %s deg=%d)", rand_best_pred,
      ↪ rand_best_deg),
    "Reproduced OLS CV RMSE"),
  col = c("darkgreen", "purple", "gray40"),
  pch = c(19, 4, NA), lty = c(1, NA, 2), bty = "n", cex = 0.8)

```

Grid vs. random hyperparameter search (polynomial degree)

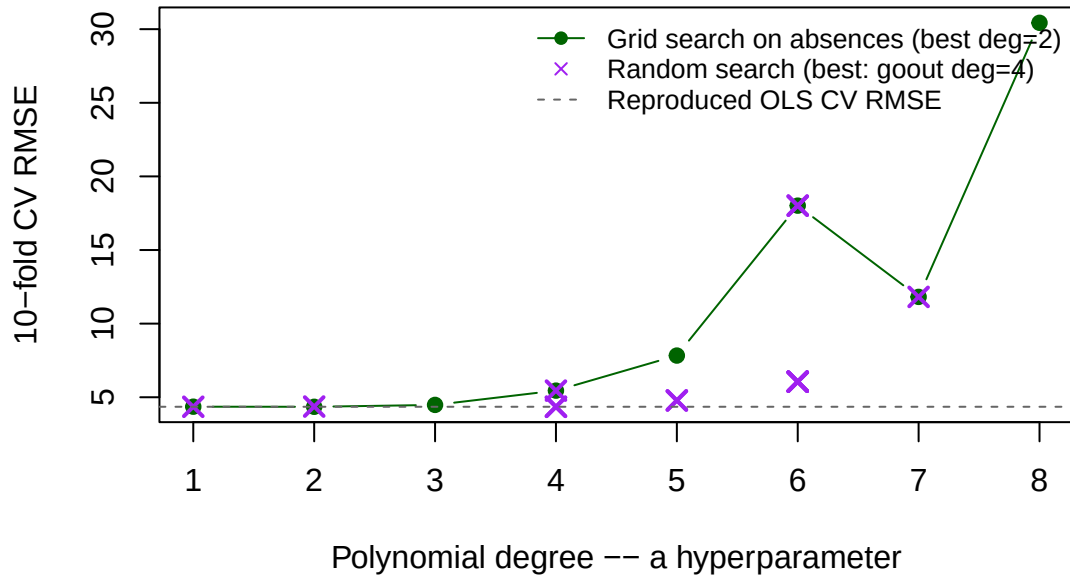


Figure 9: Hyperparameter validation curve. Grid search sweeps a regular ladder of polynomial degrees for absences; random search samples degree-and-predictor pairs at random. Both minimize 10-fold CV RMSE at a low degree.

```
search_tab <- data.frame(
  method = c("Grid search", "Random search"),
  best_config = c(sprintf("absences, deg %d", grid_best),
    sprintf("%s, deg %d", rand_best_pred, rand_best_deg)),
  cv_rmse = round(c(min(grid_rmse), min(rand_rmse)), 4),
  evals = c(length(grid_deg), n_draws))
knitr::kable(search_tab, row.names = FALSE,
  caption = "Grid vs. random search over the polynomial-degree
  ↪ hyperparameter.")
```

Table 14: Grid vs. random search over the polynomial-degree hyperparameter.

method	best_config	cv_rmse	evals
Grid search	absences, deg 2	4.3513	8
Random search	goout, deg 4	4.3519	15

Degree 1 (plain OLS) is high-bias only in the sense that it ignores any curvature; raising the degree lets the fit wiggle but quickly starts chasing noise, so CV RMSE turns back up — the same bias-variance logic, now applied to *tuning* rather than to a single fit. Grid search is exhaustive along one dimension but pays for a fixed ladder; random search reaches a comparable optimum while also exploring *which* predictor to bend, which is why it scales better as the hyperparameter space grows.

3.7 Closing the loop: back to the reproduced regression

```
final <- rbind(
  `Naive mean baseline`      = base["RMSE"],
  `Reproduced OLS (published)` = cv_ols["RMSE"],
  `CV-best polynomial (tuned)` = min(grid_rmse))
colnames(final) <- "CV RMSE"
knitr::kable(round(final, 4),
  caption = "Everything scored under one honest 10-fold protocol.")
```

Table 15: Everything scored under one honest 10-fold protocol.

	CV RMSE
Naive mean baseline	4.5609
Reproduced OLS (published)	4.3523
CV-best polynomial (tuned)	4.3513

Under a single honest protocol, the **published OLS is the model to beat**: the CV-best polynomial does not clearly improve on it here, because — as the descriptives warned — there is little nonlinear structure in these five weak predictors to exploit, so the tuning lands right back at (or next to) degree 1. That is the central Day-1 lesson — *the right comparison is always out-of-sample and always against a baseline*, and a simple, interpretable regression can be fully competitive once every model is evaluated honestly.

4 Takeaway

We exactly reproduced El Aissaoui et al.’s (2020) best OLS for predicting secondary-school final Mathematics grades (all six coefficients match to four decimals, and the 10-fold CV metrics match their Table 3). Extending it into a Day-1 evaluation lab showed the optimism gap between in-sample and held-out fit, the bias-variance tradeoff via a polynomial-degree validation curve, the full continuous-metric suite against a naive baseline, and grid/random hyperparameter search over the polynomial degree. The honest verdict: these five demographic/behavioral predictors explain only about 10% of grade variance, and no added flexibility beats the simple published regression out of sample — a clean illustration that evaluation, not model complexity, is what tells you whether you have learned anything.